

Goals-Based AI Monitoring

Concept: AI Monitor & Goals (Revised)

Aligned with [AI Monitoring & Org-Level Metrics Tech Spec](#) (April 2026)

Author	Aditya Subramaniam
Status	Draft (Revised)
Productboard	AI Monitor, AI Computed Fields, Goals & Reporting
Linear	TBD

Team

Product	Aditya Subramaniam
Engineering	Sylvia Lujo Kevin Lee Michael Meli
Design	Yvonne Doll

Consulted / Reviewed

TBD — to be populated after initial review cycle.

Changelog

Initial draft created. April 17, 2026

Revised to align with AI Monitoring & Org-Level Metrics tech spec. April 27, 2026

Business Goals

Improve admin confidence in AI automation performance by providing continuous, AI-evaluated monitoring of every AI-handled conversation, with configurable goals that track performance against measurable targets over time. This milestone introduces a closed-loop monitoring system where Quality Monitors score live conversations, Suggestions recommend improvements, Goals track business outcomes, and Anomaly Detection surfaces runtime behavioral drift.

Key Results

- Increase admin adoption of AI-powered performance monitoring (tracked via Goals, Quality Monitors, and Computed Field feature usage).
- Reduce mean time to detect AI performance degradation from days to hours.
- Achieve 80%+ admin satisfaction with AI performance visibility (measured via in-app feedback).
- Establish baseline AI Computed Field coverage across 100% of AI-handled conversations within 30 days of deployment.

Concept Summary

What we're building

Computed Fields — A new field type on the Conversation, Customer, and Company Classes whose value is produced by an LLM-based or deterministic tool. Each Computed Field has its own scope, event trigger, computation (via a tool config), and output configuration. Three default Computed Fields ship out of the box: AI-Generated CSAT (conversation-scoped, 1–5 scale), Customer Health Score (customer-scoped, composite of recent sentiment, resolution patterns, and escalation rate), and Company Health Score (company-scoped, aggregated at the company level). A 3-per-Class limit is enforced. Fields are available across the platform for routing, reporting, and automation. These fields are not seeded historically upon enable.

Quality Monitors (under Analyze) — Post-deployment evaluations that score live, closed conversations against explicit, configurable criteria. Each Quality Monitor defines multiple weighted criteria (each with its own rating categories and pass/fail determination), a pass threshold, segmentation conditions controlling which conversations are evaluated, a context profile defining what data the LLM judge receives, and an optional suggestion configuration. Two default monitors ship: an AIC AI-CSAT Monitor (if AIC is enabled) and an AIR Copilot Usage Monitor (if AIR is enabled). One additional custom monitor is permitted in M1, with room to expand as business need arises. Quality Monitors cover both AI-handled and human-assisted conversations via the Kind field ("aic", "air", or "all").

Suggestions — AI-proposed improvements generated in response to Quality Monitor failures or Anomaly Detection events. Each Suggestion targets a specific resource (a procedure, KB article, tool config, or agent), includes a proposed change with before/after content, and carries a priority (high, medium, low). Suggestions bridge the gap between observing a problem and taking action — not just showing that quality has degraded, but proposing what to change next. Admins review and apply or dismiss suggestions manually.

Anomaly Detection — Extends the existing AIC 2.0 Validation Agent into a customer-visible product surface. The Validation Agent already detects runtime deviations such as procedure non-adherence, KB non-adherence, and tool misuse. This milestone turns those internal signals into reportable anomaly events visible through a dedicated Anomaly Detection Report, and feeds them into the Suggestion pipeline so detected drift can lead to concrete improvement proposals.

Goals — Configurable performance targets tied to Computed Fields. Each goal defines a field reference, a target value, a direction (higher is better or lower is better), and a unit (percent, count, score, or currency). Admins can use three default goals (Increase CSAT, Improve Customer Sentiment, Improve Net Retention) or create custom goals referencing any Computed Field. Goals can optionally be scoped to a specific AI Agent via a Goal Resource

reference. Goals are used strictly for monitoring and alerting in this milestone, not for building or configuring AI agents.

Goal-Level Performance Reporting — A reporting surface that shows progress against each configured goal. Goal progress is tracked via point-in-time snapshots (GoalProgressSnapshot events) written each time a qualifying conversation is evaluated, capturing the field value, whether the linked evaluation passed, and the conversation context. Reporting surfaces these snapshots as trend data over time, broken down by procedure and topic where applicable, with flagged conversations shown alongside per-field scores.

Alerting — Configurable alert thresholds on Quality Monitors and Goals. Admins can opt into alerts for specific monitors and configure delivery cadence (hourly or daily). Alerts are delivered via email and in-app notifications. Alert delivery is coalesced to one notification per monitor per delivery window to prevent notification spam.

What we're not building

- Using goals to build, configure, or optimize AI agents. Goals are monitoring-only in this milestone.
- Automated remediation. When a goal is missed or a monitor fails, we alert the admin and may generate a Suggestion, but we do not automatically adjust procedures, routing, or agent behavior.
- AI Computed Fields admin UI for standalone field management. Fields are created implicitly when configuring monitors and goals. A dedicated field management UI is deferred to a future milestone.
- Custom aggregation strategies for Customer and Company level fields. M1 uses a default rolling average. Configurable aggregation (weighted average, percentile) is deferred.
- Goal-linked custom reporting queries. Tracking higher-order business outcomes such as subscription save rate, average order value, or revenue via saved reporting queries is deferred to M3.
- Monitoring all conversations. Quality Monitors evaluate AI-handled and AIR-assisted conversations. Extending to fully unassisted human conversations is planned for a future milestone.
- Kobject-scoped Computed Fields. Kobject fact pipelines require Debezium connectors and new dbt models and are deferred.

Milestone Phasing

This concept aligns with the four-phase delivery model defined in the tech spec. The concept doc focuses on the M1 customer experience but notes M0 dependencies and M2/M3 extensions for context.

M0 – Internal Foundation (Target: May 8)

- AI-Generated CSAT, Customer Health Score, Company Health Score, and general Computed Fields available internally.
- Quality Monitors, Suggestions, Anomaly Detection, and Computed Fields available internally behind internal-only controls with reasonable per-org safety caps.
- Basic cost and usage instrumentation so internal dogfooding can inform the external rollout plan.

M1 – Starter Experience + Customer Beta (Target: June 2)

- Every org that enables Kustomer AI sees a working quality monitoring experience within one click, without authoring anything themselves.
- Three pre-built Computed Fields visible to customers: AI-Generated CSAT, Customer Health Score, and Company Health Score, each with an out-of-the-box report on the reporting page.
- Beta for Quality Monitors (2 OOTB + 1 custom allowed), Suggestions, Goals, and Anomaly Detection Reports.
- Alerting available for opted-in monitors with hourly or daily cadence via email and in-app notifications.
- A “Kustomer AI Monitoring” dashboard landing page that surfaces these defaults.

M2 – Customer Authoring + Billing Limits (Target: June 15)

- Customers can author their own Computed Fields and Quality Monitors, subject to billing limits per tier/contract.
- Usage accounting and limit enforcement surface the relevant execution and cost constraints.

M3 – Goal-Linked Custom Reporting Queries (Target: July 6)

- Customers can link a custom reporting query to their goal, allowing them to track higher-level objectives such as subscription save rate, revenue, and average order value.
- Goals authored against fieldRef in M1/M2 remain backward-compatible. The integration surface is additive.

Problems & Solutions

Business Challenges

Admins deploying AI automations today lack continuous, AI-evaluated metrics. They can see conversation counts and basic resolution rates, but cannot track customer satisfaction, sentiment trends, or churn risk at the conversation and customer level.

Solution: Introduce Computed Fields and Quality Monitors to continuously evaluate every AI-handled conversation. Computed Fields write AI-derived scores (CSAT, sentiment, churn risk) onto Conversation, Customer, and Company Classes as first-class attributes available across routing, reporting, and automation. Quality Monitors score conversations against multi-criterion rubrics with weighted pass/fail evaluation. Default monitors and Computed Fields are provided out of the box so admins see value immediately upon deployment.

Admins cannot set measurable targets for AI performance or detect degradation before it impacts customers.

Solution: Goals allow admins to define a target value and direction for any Computed Field. Alerting notifies admins when a Quality Monitor detects degradation at a configured cadence (hourly or daily). The reporting surface shows real-time progress against each goal, making it easy to spot trends and take action before metrics deteriorate further.

When problems are detected, admins have no systematic path from observation to action.

Solution: Suggestions are auto-generated when Quality Monitors fail or Anomaly Detection events fire, proposing concrete changes to procedures, KB articles, tool configs, or agent settings. Each suggestion includes before/after content and a rationale, so admins can review and apply fixes without starting from scratch.

Technical Challenges

Computed Fields must be evaluated on every AI-handled conversation without meaningfully impacting conversation latency or throughput.

Solution: Evaluate asynchronously post-conversation so the customer experience is never impacted. Implement queue-based processing with backpressure handling and debouncing to prevent redundant evaluations from automations that repeatedly open/close conversations.

Quality Monitor results generate high data volumes that must be retained for both operational access and long-term analytics.

Solution: Dual-store architecture. MongoDB retains operational data with a 30-day TTL on evaluation results (conversationevaluationresults collection) for recent drill-down access. ClickHouse retains analytical data for 24+ months via staging, intermediate, and mart

models. After 30 days, the UI falls back to ClickHouse summary data (evaluation outcome, score, pass/fail) without per-criterion rationale detail.

Jobs To Be Done

ID	PERSO NA	JOB	SUCCESS INDICATOR
JOB 1	Admin	When deploying an AI automation, I want to define measurable performance goals, so that I can track whether my AI is meeting my standards over time.	Admin can configure goals tied to Computed Fields with target values, directions, and units, and can do so using defaults or custom goals, without needing to write queries or build reports manually.
JOB 2	Admin	When monitoring AI performance, I want AI to automatically score every AI-handled conversation on metrics like CSAT, sentiment, and churn risk, so that I have continuous, objective data without needing to manually review conversations.	Quality Monitors evaluate every qualifying conversation and write scores to Computed Fields on the Conversation and Customer Classes automatically without admin intervention after initial configuration.
JOB 3	Admin	When a performance metric degrades, I want to be alerted at a cadence I choose, so that I can investigate and fix issues before they impact more customers.	Admin receives alerts via email or in-app at hourly or daily cadence when an opted-in Quality Monitor detects degradation, without needing to manually check the dashboard.
JOB 4	Admin	When reviewing AI performance, I want to see goal-level reporting broken down by procedure, topic, and time period, so that I can pinpoint exactly where my AI is underperforming.	Admin can view reporting with per-goal trend data, procedure and topic breakdowns, and flagged conversations from a single view without switching between multiple reports.
JOB 5	Admin	When investigating flagged conversations, I want to see the per-criterion Quality Monitor scores alongside the reason it was flagged, so that I can quickly understand what went wrong.	Admin can see per-criterion scores, weights, pass/fail status, and rationale for each conversation with a single click to view the full conversation.
JOB 6	Admin	When a Quality Monitor flags a failure, I want an AI-generated Suggestion that proposes a concrete fix, so that I can act on problems instead of just observing them.	Admin receives Suggestions targeting a specific procedure, KB article, or tool config with a proposed change (before/after), rationale, and priority, and can apply or dismiss from the Suggestions view.
JOB 7	Admin	When my AI agent deviates from expected behavior at runtime, I want to see those	Anomaly Detection events from the AIC 2.0 Validation Agent are visible in

		anomalies surfaced in a report and linked to improvement suggestions, so that I can catch drift before it becomes a pattern.	a dedicated Anomaly Detection Report and feed into the Suggestion pipeline.
JOB 8	Admin	When I need to track a custom metric specific to my business, I want to add a new goal tied to a Computed Field, so that I can monitor what matters most to my organization.	Admin can create custom goals by specifying a Computed Field reference, target, direction, and unit from the Goals configuration screen without needing engineering support.
JOB 9	Admin	When configuring Quality Monitors, I want to define weighted criteria with rating categories and a pass threshold, so that the scoring matches my business context and priorities.	Admin can add, weight, and mark criteria as essential, configure rating categories per criterion, and set a pass threshold from the Quality Monitor configuration screen.

Initial Spec Ideas

Exploratory first thoughts to pressure-test feasibility and direction. Non-binding and subject to change after review.

Computed Fields

Computed Fields are AI-generated or deterministic attributes written back onto Kustomer resources (conversations, customers, companies). They are standalone product primitives with their own scope, event trigger, computation pipeline, and output configuration.

Default Computed Fields

- **AI-Generated CSAT** — Conversation-scoped. LLM-inferred CSAT from a closed conversation's transcript, written to a conversation custom field on a 1.0–5.0 scale.
- **Customer Health Score** — Customer-scoped. LLM-graded composite of recent conversation sentiment, resolution patterns, and escalation rate over the last 90 days. Rolling average on the Customer Klass.
- **Company Health Score** — Company-scoped. Same pattern aggregated at the company level.

Computed Field Configuration

- **Name** — Friendly label for the field.
- **Description** — What the field measures and why it matters.
- **Scope** — Which resource type the field lives on: Conversation, Customer, Company, or Kobject (Kobject deferred).
- **Trigger** — The event that initiates computation (e.g., `kustomer.conversation.update:done`).
- **Computation** — References a tool config that defines how the field is calculated (LLM-based or deterministic).
- **Output** — The custom field key the value is written to, the value type, and an optional rationale field name for AI explanations.

Limits and Constraints

- 3 Computed Fields per Klass (each default counts toward the limit).
- Fields are not seeded historically upon enable.
- Evaluation is asynchronous post-conversation (fire-and-forget with observability).
- Debouncing: if multiple trigger events fire for the same resource within a short window, only the last is processed.
- Optimistic locking with PATCH operations on subjects, retrying on conflict.

Quality Monitors (Analyze > Quality Monitors)

Quality Monitors are post-deployment evaluations that score live, closed conversations against explicit criteria. They are the core operator-facing surface for understanding whether AI-handled and human-assisted conversations are behaving as intended.

Default Monitors

- **AIC AI-CSAT Monitor** (if AIC is enabled) — Evaluates AI-handled conversations for resolution quality, accuracy, tone, and customer satisfaction signals.
- **AIR Copilot Usage Monitor** (if AIR is enabled) — Evaluates human-assisted conversations for copilot adoption, suggestion acceptance, and quality impact.

Monitor Configuration

- **Name / Description** — Friendly label and explanation of what the monitor evaluates.
- **Kind** — Which conversations are evaluated: "aic" (AI-handled), "air" (human-assisted), or "all".
- **Segmentation** — Conditions (structured) and/or natural-language conditions controlling which conversations match.
- **Context Profile** — Controls what data the LLM judge receives: messages, automation config, execution traces, copilot messages, relevant KB articles, customer data, and/or kobject data.
- **Analysis** — The scoring rubric:
 - **Criteria** — Each criterion has a name, type ("ai", "tool", or "condition"), instructions or tool config reference, rating categories (each with a name, description, pass/fail flag, and score percentage), a weight (relative importance), and an essential flag (failing an essential criterion = entire evaluation score becomes 0%).
 - **Pass Threshold** — A 0.0–1.0 threshold that the overall weighted score must meet for the conversation to pass.
- **Output Field** — Optional fields where the overall score and pass/fail result are written as Computed Fields on the conversation.
- **Suggestion Config** — Whether suggestions are enabled for this monitor, and which target types are eligible (procedure, KB article, tool config, agent).
- **Goal IDs** — Links this monitor to one or more Goals for progress tracking.

One Custom Monitor in M1

Beyond the two defaults, admins can create one additional custom monitor in M1. This allows admins to define their own criteria, weights, and thresholds for a use case specific to their business. The limit can be expanded as business need arises.

Suggestions

Suggestions are AI-proposed improvements generated when a Quality Monitor fails or an Anomaly Detection event fires. They are paired with monitors and anomalies rather than positioned as a standalone product surface.

Suggestion Fields

- **Type** — What kind of change: `procedure_edit`, `kb_update`, `tool_config`, or `agent`.
- **Status** — pending, applied, or dismissed.
- **Priority** — high, medium, or low.
- **Target** — The specific resource the suggestion applies to (resource type, ID, and name).
- **Proposed Change** — Change type (edit or create), with before and after content so admins can see exactly what would change.
- **Rationale** — Explanation of why the change is recommended, linked to the Quality Monitor failure or anomaly that triggered it.

Admins review suggestions and apply or dismiss them manually. Dismissed suggestions require a dismiss reason. Repeated suggestions for the same target are deduped within a bounded time window.

Anomaly Detection

Anomaly Detection extends the AIC 2.0 Validation Agent (already shipped in production) into a customer-visible surface. The Validation Agent inspects an agent's in-flight or just-completed reasoning and flags deviations such as procedure non-adherence, KB non-adherence, and tool misuse.

- **Anomaly events** are emitted with an anomaly type, description, trigger source, and affected resource.
- **Anomaly Detection Report** surfaces these events in a dedicated reporting view.
- **Suggestion pipeline integration** — Anomalies feed into the same Suggestion pipeline as Quality Monitor failures, generating proposed fixes when actionable.

Goals (Settings > Goals)

Goals are configurable performance targets that reference Computed Fields. Each goal defines what metric to track, a target value, and the direction that indicates improvement.

Default Goals

- **Increase CSAT** — References AI-Generated CSAT Computed Field. Direction: higher is better. Target: configurable (suggested default 4.5).
- **Improve Customer Sentiment** — References Customer Health Score Computed Field. Direction: higher is better.
- **Improve Net Retention** — References a churn risk Computed Field. Direction: lower is better.

Goal Configuration Fields

- **Goal Name** — Short label (e.g., "Improve CSAT").
- **Description** — Natural language objective.
- **Field Refs** — One or more Computed Field paths tracked by this goal.
- **Goal Resource** (optional) — Scopes the goal to a specific resource type and ID (e.g., a specific AI Agent/automation).
- **Metric** — The primary tracked field, consisting of:
 - **Field Ref** — Must match one entry in Field Refs.
 - **Target** — Numeric target value.
 - **Direction** — "higher_is_better" or "lower_is_better".
 - **Unit** — "percent", "count", "score", or "currency".
- **Evaluation IDs** — Links to Quality Monitors whose results contribute to goal progress tracking.

Goal Progress Tracking

Goal progress is tracked via GoalProgressSnapshot events. Each snapshot captures the goal ID, field reference, current field value, the conversation and evaluation that produced it, and whether the evaluation passed. These snapshots are aggregated daily in ClickHouse (avg/min/max values, pass/fail counts) and surfaced in the reporting view as trend data.

Open Questions

- Do we need a maximum number of goals per org, or is this unlimited?
- Should Goals support being scoped to multiple automations, or is one-to-one sufficient for M1?

- How do we handle the case where an admin creates a goal referencing a Computed Field that no monitor writes to? Should we warn them, or allow it and show "no data"?

Alerting

Alerting allows admins to opt into notifications when Quality Monitors detect degradation.

Alert Configuration

- **Opt-in per monitor** – Admins choose which monitors trigger alerts.
- **Cadence** – Hourly or daily.
- **Delivery channels** – Email and in-app notifications.
- **Coalescing** – One notification per monitor per delivery window. Delivery retries are rate-limited to avoid notification spam.

Open Questions

- Should alerts support delivery to specific users or teams, or is org-admin-only sufficient for M1?
- Should goal-level alerts (not just monitor-level) be supported in M1, or deferred?

Reporting Surfaces

The following reporting surfaces ship in M1:

- **AI-Generated CSAT Report** – OOTB report on the reporting page for the AI-CSAT Computed Field.
- **Customer Health Score Report** – OOTB report for the customer-scoped health score.
- **Company Health Score Report** – OOTB report for the company-scoped health score.
- **Anomaly Detection Report** – Surfaces AIC 2.0 Validation Agent anomaly events.
- **Kustomer AI Monitoring Dashboard** – Landing page surfacing default monitors, goals, recent alerts, and flagged conversations.

Goal-level trend data (from GoalProgressSnapshot events) is available through the monitoring dashboard. Detailed procedure-level and topic-level breakdowns depend on the data available in ClickHouse intermediate models (evaluation performance hourly, goal progress daily).

Data Retention

Quality Monitor results are stored in MongoDB with a 30-day TTL on the conversation evaluation results collection. For conversations older than 30 days, the UI falls back to

ClickHouse summary data (evaluation outcome, score, pass/fail) without per-criterion rationale detail. ClickHouse retains analytical data for 24+ months. Suggestions retain a 30-day TTL past resolution; pending suggestions never expire via TTL.

Existing Screens (Preserved)

The following existing screens are preserved as-is within the AI for Customers navigation and are not modified in this milestone:

- **Build** – Guidance, Knowledge Sources, and Procedures configuration.
- **Test** – Test categories and evaluation scores (pre-deployment Evaluations, distinct from Quality Monitors).
- **Deploy** – Conditions, Smart Routing, and Review Evaluations.
- **Settings > General** – AI Automation Settings, Guardrails, Handoff, Conversation Start, Verification, Abandonment, Fallback, and Business Hours.
- **Analyze > Observability** – Traces and execution logs.

Out of Scope

- **Goal-driven AI agent optimization:** Using goals to automatically adjust AI agent behavior, procedures, or routing. Goals are monitoring-only in M1.
- **Automated remediation:** When a goal is missed, we alert the admin and may generate a Suggestion, but do not automatically fix the issue.
- **AI Computed Fields admin UI:** A dedicated screen for managing (creating, renaming, deleting) Computed Fields is deferred. In M1, fields are created implicitly via monitor and goal configuration.
- **Custom aggregation strategies:** Configurable aggregation for Customer and Company level fields is deferred. M1 uses a default rolling average.
- **Goal-linked custom reporting queries:** Tracking higher-order business outcomes via saved reporting queries is deferred to M3. Goals in M1/M2 use fieldRef; M3 will add a queryId or similar reference additively.
- **Kobject-scoped Computed Fields:** Requires Debezium connector and new dbt models. Deferred.
- **Historical backfill:** Computed Fields are not seeded historically upon enable.
- **User-authored raw SQL:** Authoring happens through structured config, not free-form expressions.

Risks & Mitigation

Evaluation latency and throughput: Running AI evaluations on every conversation could introduce latency or create a processing bottleneck at scale.

Mitigation: Evaluate asynchronously post-conversation. Implement debouncing so repeated trigger events for the same resource within a short window only process the last one. Start with the default monitors and measure throughput before expanding custom monitor limits.

LLM cost runaway: One enterprise org enabling several Computed Fields and monitors across high-volume traffic could cost far more than expected.

Mitigation: Per-org daily execution caps. 3-per-Klass Computed Field limit. Monitor limits (2 OOTB + 1 custom in M1). Cost alerting when org crosses thresholds. Billing-based enforcement before broad M2 rollout.

Concurrent automation conflicts: Computed Fields introduce another automation system running alongside Workflows, Business Rules, and Queues/Routing, increasing the risk of concurrent writes to the same resource.

Mitigation: Optimistic locking with retry on conflict for Computed Field PATCH operations. Batching multiple Computed Fields for the same Klass and trigger into a single PATCH where possible.

AI Computed Field accuracy: AI-generated scores may not always align with ground truth, especially for nuanced conversations.

Mitigation: Allow admins to customize evaluation criteria and prompts. Surface flagged conversations with full per-criterion scores so admins can audit AI evaluations. Consider adding a human feedback loop (thumbs up/down on AI scores) in a future milestone.

30-day TTL impact on drill-down: After MongoDB TTL expiration, per-criterion rationale detail is unavailable.

Mitigation: Fall back to ClickHouse summary data (outcome, score, pass/fail). Document this limitation clearly. Evaluate whether mirroring rationale to a cheaper store is warranted based on customer feedback.

Alert fatigue: Poorly tuned monitors or repeated failures could produce noisy notifications.

Mitigation: Opt-in alerts only. One notification per monitor per delivery window. Delivery throttles. Easy mute/disable controls.

Scope creep into agent optimization: Stakeholders may push to use goals for automated agent tuning in M1.

Mitigation: Clearly scope M1 as monitoring-only. Goals power dashboards, alerts, and suggestions, not agent behavior. Document the vision for goal-driven optimization so stakeholders know it's planned, just not in this milestone.

Competitor Analysis

See the separate [Competitive Research document](#) for a full landscape analysis. Key areas evaluated for this initiative include how competitors (Intercom, Zendesk, Sierra, Decagon, Siena, Gorgias) surface AI performance metrics to admins, whether they offer configurable goals or targets, how they handle AI-evaluated quality scoring, and what alerting mechanisms exist for AI performance degradation. The core finding remains: no competitor has shipped a true goals-first configuration experience where admins declare business outcomes and the platform auto-steers monitoring, evals, and reporting. Quality Monitors, Computed Fields, and Goals together position Kustomer to claim this whitespace.

-